

SALONI DEEPAK

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EDUCATION

Ph.D. in Physics (Anticipated August 2027) <i>University of Notre Dame</i> Thesis title: The Metal Enrichment of Diffuse Gas at the Peak of Cosmic Star Formation PhD supervisors: J. C. Howk, Nicolas Lehner	Notre Dame, IN, USA Spring 2021–present
M.S. Physics <i>University of Notre Dame</i>	Notre Dame, IN, USA 2024
M.Sc. in Physics & Astrophysics <i>University of Delhi</i>	New Delhi, India 2018–2020
B.Sc. Honours in Physics <i>St. Xavier's College</i>	Kolkata, India 2015–2018

AWARDS & DISTINCTIONS

Future Investigators in NASA Earth and Space Science and Technology (FINESST) award (\$150,000), National Aeronautics and Space Administration (NASA)	2024–2027
American Astronomical Society International Travel Grant	2024, 2026
GSG Conference Presentation Grant , University of Notre Dame	2024, 2026A, B
Graduate School Professional Development Award (GSPDA) : Zahm Professional Development Fund, University of Notre Dame	2026
GSPDA : Notebaert Professional Development Fund, University of Notre Dame	2024, 2026
Ashish Palit Memorial Prize, Awarded for graduating with the highest marks in B.Sc. Physics Honours, St. Xavier's College, Kolkata	2018
INSPIRE, Scholarship for Higher Education (SHE) Scholar, the Department of Science and Technology, Ministry of Science Technology, Government of India	2015–2020

GRANTS & RESEARCH AWARDS

Co-I: "Illuminating the Dark Ages of Metal Evolution: An HST Legacy Survey at Cosmic Noon," Hubble Space Telescope (HST), Cycle 32, AR 17862	2024
Co-I: "Metals at Cosmic Noon," Astrophysics Data Analysis Program, NASA ADAP24-0169	2024
Co-I: "A Complete Characterization of Multiphase Circumgalactic Gas in the Cosmic Afternoon through Forward Modeling," HST, Cycle 33, ID 18146	2025
Co-I: "Ground Truth for the CGM: A Cloud-by-cloud Ionization Modeling Challenge with pc-scale CGM simulations," HST, Cycle 34, ID 18465	2025

PUBLICATIONS

1. *A Global Census of Metals in the Universe*, **Deepak, S.**, Howk, J.C., Lehner, N., Péroux, C., (2025) *ApJ*, Volume 987, Issue 2, id.199, 24 pp
2. *The BRIDGE Survey. I. Metallicities and Physical Properties of Gas probing the Circumgalactic Medium at Cosmic Noon*, **Deepak, S.**, Lehner, N., Howk, J.C., O'Meara, J., Péroux, C., Prochaska, J.X., (2026) (*submitted to ApJ*) [\[website\]](#)
3. *The Red Dead Redemption Survey of Circumgalactic Gas About Massive Galaxies. II. Low-metallicity Gas is Not Rare in the Inner Halo*, Berg, M. et al (including **Deepak, S.**) (2026), (*submitted to ApJ*)

4. *Transverse Metallicity Variations on Sub-kpc Scales toward the Lensed Quasar HE1104–1805*, **Deepak, S.**, Lehner, N., Howk, J.C., López, S., Cortés, H., (2026) (*in prep, to be submitted October 2026*)

TALKS, POSTERS & SEMINARS

Invited Talk: “The BRIDGE survey: Metallicities and Physical Properties of the CGM” <i>The Milky Way Halo Group, Space Telescope and Science Institute, (STScI), Baltimore</i>	2026
Talk: “Bridging the gap at Cosmic Noon: Metals in the CGM” <i>Figuring out Gas and Galaxies in Enzo (FOGGIE) meeting, STScI, Baltimore</i>	2026
Invited Talk: “Bridging the gap at Cosmic Noon: Metals in the CGM” <i>Galactic Metal Cycles, University of Notre Dame, Kylemore, Ireland</i>	2026
Talk: “BRIDGE-I: Metals in the CGM at Cosmic Noon” <i>Celebrating 40 years of Damped Lyman alpha Absorbers, Kavli IPMU, University of Tokyo</i>	2026
Talk: “Towards a Global Metal Budget of the Universe: the missing link at Cosmic Noon” <i>31st Annual Keck Science Meeting, University of California, Los Angeles</i>	2025
Seminar: “Towards a Global Metal Budget of the Universe” <i>Department of Physics and Astronomy, University of Notre Dame</i>	2025
Talk: “A global census of metals in the Universe” <i>Baryons Beyond Galactic Boundaries, The Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, India</i>	2024
Talk: “The global distribution of metals in the Universe” <i>FOGGIE meeting, Michigan State University</i>	2024
Poster: “A global census of metals in the Universe” <i>Oases in the Cosmic Desert: Understanding the Structure of the Circumgalactic Medium, School of Earth and Space Exploration, Arizona State University</i>	2023
Talk: “Determining the host luminosity of a gravitationally lensed AGN” <i>Visiting Student Research Program, IUCAA, Pune, India</i>	2019

WORKSHOPS & SCHOOLS

AstroAI workshop, Center for Astrophysics (CfA), Harvard & Smithsonian	2025
Code/Astro , CIERA, Northwestern University	
(Developed and packaged absorption line finder tool pyALF)	2024
Michigan Cosmology Summer School , University of Michigan, Ann Arbor	2023
Jupyter notebook workshop , Developing accessibility tools in Astronomy on the “Day of Accessibility”, Space Telescope Science Institute (STScI)	2023
Summer School in Statistics for Astronomers XVII , Penn State University	2022
Visitor Student Research Programme, IUCAA, Pune, India	2019
Research student at National Initiative on Undergraduate Science , HBSCE–Tata Institute for Fundamental Research (TIFR)	2016–2017

OTHER RESEARCH EXPERIENCE

Polytropic gas atmosphere around a spherically symmetric gravitating core Research Advisor: Prof. T. R. Seshadri, University of Delhi, New Delhi, India	2020
M.Sc. Thesis: Magnetic field of the Milky Way from Faraday rotation of pulsar signals Thesis supervisor: Prof. T. R. Seshadri, University of Delhi, India	2019–2020
Determining the host luminosity of a gravitationally lensed AGN Research Advisor: Dr. Anupreeta More, IUCAA, India	2019
B. Sc. Thesis: The Dynamics of Spherical Bondi Accretion: An Overview	

Thesis supervisor: Prof. Suparna RoyChowdhury, St. Xavier's College, India	2018
Estimation of extent of convective cores in pulsating stars	
Research Advisor: Prof. Anwesh Mazumdar, TIFR, India	2016–2017

APPOINTMENTS

Research Assistant	Notre Dame, IN
<i>University of Notre Dame</i>	Fall 2021–present
Teaching Assistant for Intermediate Mechanics	Notre Dame, IN
<i>University of Notre Dame</i>	Spring 2021

MENTORING & SERVICES

Mentorship program for Undergrads, Department of Physics and Astronomy, Notre Dame Mentee:	
Anandini Mitra	2024–2026
Judge at the Northern Indiana Regional Science and Engineering Fair (NIRSEF)	2024
Chair of International Committee, Graduate Physics and Astronomy Society, Notre Dame	2023–2024
Graduate Physics and Astronomy Society Mentorship program, University of Notre Dame	2023
Mentorship for primary school education, the National Service Scheme, the Ministry of Youth Affairs and Sports, Government of India	2015–2018

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, FORTRAN
Astronomical Software: MESA (stellar modeling), glafic (gravitational lens modeling), SAOImage DS9, Cloudy (spectral analysis and ionization modeling), Pypeit (spectra calibration and reduction)
Data and Code Management: GitHub (<https://github.com/sdeepak1997>)

REFERENCES

Dr. J. Christopher Howk Professor Department of Physics and Astronomy University of Notre Dame jhowk@nd.edu	Dr. J. Xavier Prochaska Professor Department of Astronomy and Astrophysics UCO/Lick Observatory, University of California xavier@ucolick.org
Dr. Nicolas Lehner Research Professor Department of Physics and Astronomy University of Notre Dame nlehner@nd.edu	Dr. Evan Kirby Professor Department of Physics and Astronomy University of Notre Dame ekirby@nd.edu